Cognitive Psychology of Attention- PS20167

Types of attention

- Attention can be driven from within based on our current goals and desires (e.g. getting a high grade in unit)
- Or it can also be attracted by certain events in the environment around us (e.g. a loud noise in the hallway)
- These are commonly referred to as voluntary vs reflexive attention, or also endogenous vs exogenous attention

Voluntary (endogenous) attention

- Our ability to intentionally attend to something
- It's a top-down and goal-directed process
- Hopefully your attention is focused on me and my material to some degree during the lecture
    - There are many more interesting things going on right now that you could be doing
    - But this material is important for your goals of doing well on the exam and this Unit – so you focus
- I like to refer to this as our attention being 'pushed'

Reflexive (exogenous) attention

- But attention can also be attracted by certain information or events in our environment
- It's a bottom-up stimulus-driven process
- If there was a gun shot in the hallway everyone would turn and pay attention to it
    - Your attention would be grabbed or pulled by information happening around you
    - Doesn't matter your internal goals
- I like to refer to this as our attention being 'pulled'

Attention "is the taking possession of the mind, in clear and vivid form, of one out of what may seem several simultaneously possible objects or trains of thoughts…" (William James)

Exogenous attention in research

Posner cueing tasks

Valid trial

- Your attention is cued to a location, and then a target appears there
- Target appears at same location of the cue

Invalid trial

- Your attention is cued to a location, and then a target appears on the opposite side

Neutral trial

- Your attention is cued, but not to a specific location, so the cues were not informative towards the location of the target

No cue trial

- Your attention wasn't cued to a specific location, so nothing informative appeared about the location of the target

People are faster to respond to target when it appears in the same location as the cue – better at processing information in the location they are attending to

Endogenous attention in research

Valid trial

- Your attention was cued to a location e.g. with an arrow, and then the target (dot) appeared there

Other valid trials

- People are faster and more accurate on these types of trials
- It's not just non-social cues shifting our attention
    - these attentional processes going on all around us…

Social cues e.g. faces also push attention
Cue pushes attention a certain way e.g. through an arrow – this may not be exactly synonymous with people's interests or goals/endogenous attention
People faster on valid trials

Dot Probe Task

- In the dot probe task 2 pictures are presented side by side and then dots appear in location of one of the pics
- Participants must press a button for the dots
    - often it's the orientation of the dots (**:** or **..**)
    - In the slides that follow you have to determine whether the dots are horizontal or vertical

- People are generally faster to make decision about target/dots when they appear behind location where the faces were presented
    - your attention was drawn more to the faces
    - so more of your attention was on the dots
    - so, you are faster to make response about them
- this infers that people have a natural interest in people which directs them to look at them
- Interestingly, people with autism don't show this same effect for faces – reduced automatic orienting to faces

"It implies withdrawal from some things in order to deal effectively with others" (William James)

Attention is limited

- Focusing attention means information is processed better within that focus….
- but information is processed *less* outside that focus

- so, items are easily missed

Change blindness experiments
- attention needs to be focused on the location in order to successfully spot the change
- 2 identical pictures are shown after one another repeatedly
- There is one item that is changed between pictures, either colour, location, or absence
- You need to spot which item is changed
- A visual disruption is included between the 2 images being shown which kind of wipes out the processing for a brief moment – makes it harder to spot the change unless you are looking at it

Section summary
- Attention is a concentration of mental activity, can be from within or from environment
- Attending to one item/area means less processing of information outside that area
    - means it's easy to miss information that is outside your spotlight of attention

Models of attention

Dichotic Listening
- Cherry (1'5–): "cocktail party effect"
    - Cocktail party effect – can easily pay attention to one conversation even if there are many other conversations occurring in the room at the same time.
    - You can also easily focus
- Dichotic listening task
    - Shadowing
    - Participants mostly attended to the physical properties of the message
    - Were unaware if the speaker changed from male to female or changed language form English to German.

Broadbent's (1'5') Filter Model
- Early-selection model
- One of the first information-processing models
- Sees the attention system as a series of channels
- Information first processed by specialised channels
    - e.g. acoustic or visual channels
    - These channels operate simultaneously
- Information is then brought into a buffer store
    - A filter chooses the stimuli for further processing
    - Selection is made based on physical properties
    - The chosen stimuli become the focus of attention
    - The other stimuli are completely disregarded

Evaluation of Broadbent's model

- Takes account of Cherry's findings
- BUT… what about when intrusion from semantically salient stimuli? (Moray, 1'5'; Underwood, 1'74)
- E.g. when you're in a busy pub attending to a conversation but someone shouts your name, you will attend to it – certain types of information can intrude attention which wouldn't be possible if we stopped processing it/early on the information had been filtered out, blocked, or not processed any further

<u>Treisman's (1'65) Attenuator Model</u>

- Early-selection model improving on Broadbent's filter model
- Information is processed and continues being processed but the filter turns the volume down on different types of stimuli – the stimuli we are most interested in and attending to get a high volume, and others get a lower volume or almost no volume if it's really unimportant – so it attenuates
- If it has a low volume, we still can process it and get into our consciousness but is just less likely to
- Attenuators process stimuli to different degrees
    - This is unlike in the filter model where stimuli are processed in an all-or-nothing fashion
    - This allows for simultaneous processing of stimuli from different channels
- Selection is based upon physical characteristics but also semantic relevance and familiarity
- Physical properties still capture and hold attention
    - However, the attenuators are also influenced by previously analysed material
    - e.g. highly familiar or stimuli semantically relevant in the context

Evaluation of Treisman's model

- Takes account of intrusion of salient information
- But how is information selected for processing?
- Assumes unselected information is left to decay in sensory memory – that it is processed so much less that is isn't going to go into our memory
    - Therefore, it will have no effect on further behaviour
    - Is this hypothesis valid?

<u>Late-Selection Model</u>

- Deutsch … Deutsch (1'6–); Norman (1'6')
- Less evidence for these sorts of models
- Selection happens at the level of memory
    - Expectations are set up dependent upon context of the story

Evidence

- Having something you've seen before can affect if you are going to process / recall it later if you didn't process / attend to it / go to your memory

Theory of Perceptual Load

- Lavie (1''5; 2000)
- Hard to compare with early filter models
- Instead of being seen as a filter, attention is seen as 'a system' with limited resources
- How much attention resources does the task demand? If something is difficult, it will take up more of attention and focus to be able to do it
- Think about learning to drive (e.g. initially think about everything going on but after some time processes become automatic)

Evidence for Lavie's model

- Experiments of attentional load
- Respond to the following trials about whether the target is a lowercase "x" or "z"
- High versus low load
    - You probably noticed that the first trial was easier, and the second trial was harder (OOXOOO)
    - The distractors were much different to target, so little attention resource used up in task (NMSKXV)
    - The distractors similar to targets, therefore more attentional resources needed for task
- Under low load attention is influenced by the incompatible distractors – performance differs
- Under high load there are little attentional resources available – no difference in performance because the distractors are not being 'processed' in same way – more attention needed for the task meant little attention on the distractor

Strengths of theory of perceptual load

- It doesn't require a filter process – therefore it does not involve such a rigid system
- See effects with factors associated with task
    - e.g. familiarity, motivation, ability, personality traits

What is attention?

- William James (1''0): "Everybody knows what attention is"
- Matlin (2005): Attention is a *concentration of mental activity*
- We are rarely processing just one object at a time without processing anything else as in experiments
- We typically are immersed and view complex scenes with lots of potential targets and distractor to attend

Multitasking

- Driving
    - Driving is a complex task requiring the coordination of many different mental and physical processes
    - Attention needs to be efficiently directed between them
    - Very little in our evolution has prepared us for reacting to split-second events in a half-ton metal box while moving at rapid speed
    - A common cause of car crashes
        - Dingus et al. (2006) - 100-Car Naturalistic Driving Study
        - Video-recorders in 100 cars
        - Recorded 2 million miles of driving
        - ‘2 crashes
        - 771 near-misses
        - 22% involved using a mobile phone
    - Using a mobile phone
        - Using a mobile phone disrupts cognitive processes involved in driving (Drews et al., 200‘; Strayer et al., 200–)
        - Level of impairment has been likened to driving with a blood alcohol level of .0‘ (current UK limit)
        - Four times more likely to have an accident if using a mobile phone - using hands-free makes no difference (Redelmeier … Tibshirani, 1’’7)
        - Why does it cause such a problem?
          → could this be due to the overloading of attentional resources?
          → how can we test this
        - Is it due to the holding of the phone? I.e. a motor impairment from using the phone itself disrupting the driving of the car? → NO!
          → Reaction to red/green lights in a driving simulator was impaired when drivers communicated via either a hand-held or hands-free device (Strayer … Johnston, 2001)
        - Driving and conversing takes up too much of our attentional resources? → NO!
          → Having a conversation is not the problem as having a passenger in the car has been found to reduce the risk of accident (Drews et al., 200‘)
    - Attentional disruption while driving
        - Drews et al. (200‘) - advanced driving simulation
        - Driver spoke with a friend either in the simulator or remotely via a mobile phone
        - Driving performance was significantly worse when speaking on the phone vs a passenger
        - Increased lane drift
        - Vehicle distance affected
        - Navigation ability affected
        - The passenger can view the road and therefore is sensitive to the driving conditions
        - As conditions become more demanding:

        1) drivers with a passenger reduced their speech rate … passengers modified speaking to match road conditions
        2) those in the 'phone-condition' increased their speech rate (to fill gaps in conversation?) - they couldn't see the road so didn't modify their speech
    - Speaking via a mobile phone competes for the very limited attention resources we have
    - Telephone use is a task for which the cognitive system did not evolve to deal with
    - Telephone use while driving negatively affects how behaviour and attention are co-ordinated

<u>Section summary</u>

- Driving while using a mobile phone disrupts driving ability even when using hands-free
- Most likely due to the cognitive demands of using the phone